Case Study

Integrating Analytical Hierarchy Process and Machine Learning for Enhancing Flood Hazard Mapping

Nikunj K. Mangukiya, S.M.ASCE[1]; Anupal Baruah[2]; Diganta Barman[3]; B. M. Arjun[4]; and Ashutosh Sharma[5]

Downloaded from ascelibrary.org by UFRPE - UNIVERSIDADE FEDERAL RURAL DE PERNAMBUCO on 09/24/26. Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

Abstract: Floods, intensified by climate change and anthropogenic activities, pose significant global threats to human life and infrastructure. Although several studies have demonstrated the efficacy of the multicriteria decision-making (MCDM) methods involving human expertise and machine learning (ML) methods utilizing machine intelligence for developing flood hazard maps, there is a need for an integrated approach that combines the advantages of both methods and addresses their respective shortcomings. This study proposes an integrated MCDM and ML framework using the analytical hierarchy process and the random forest algorithm (AHP-RF). The upper Assam region of India, highly susceptible to floods, was selected as a case study. Factors influencing flooding, such as topography, land cover, embankment breach, river confluence points, normalized difference moisture, vegetation and water index, and rainfall-runoff characteristics, were utilized for developing a high-resolution flood hazard map using the proposed AHP-RF approach. The predicted maps were compared with historical flood inventory utilizing the area under the curve (AUC) and accuracy metrics for validation. Results indicate that the AHP-RF approach provides more accurate representations of flood hazard zones, with an improved AUC score of 0.74 and accuracy of 0.792 compared to the standalone AHP (AUC score of 0.654 and accuracy of 0.595). The derived high-resolution flood hazard map highlights that the built-up and agricultural lands of the Dhemaji and Lakhimpur districts are highly susceptible, while Dibrugarh and Tinsukia are moderate and low susceptible, respectively. Overall, the proposed AHP-RF approach can offer a reliable flood hazard map, aiding authorities and stakeholders in flood mitigation. **DOI: 10.1061/NHREFO.NHENG-2188.** *© 2025 American Society of Civil Engineers.*

Author keywords: Analytical hierarchy process (AHP); Flood hazard; Machine learning; Random forest; Upper Assam.

Introduction

Floods, being natural calamities, have a global impact, causing extensive damage to infrastructure, disrupting economic stability, and endangering societal means of living (Mohapatra and Singh 2003; Yadav and Mangukiya 2021). In emerging economies such as India, where flood-prone areas cover approximately 12.5% of the country's areas (Mohanty et al. 2020), floods have an enormous impact due to unplanned urbanization, dam overflowing in conjunction with extreme local precipitation, and encroachments of floodplain (Baruah et al. 2023; Beshir and Song 2021; Mangukiya and Yadav 2022; Singh and Kumar 2017). Considering the anticipated rise in flood frequency and intensity as well as the severity of extreme events due to global climate change, it is imperative to identify flood hazard zones for informed decision-making on preventive strategies (Ali et al. 2019; Thakur and Mohanty 2023).

The term "flood hazard" is commonly understood as the likelihood of a flood occurrence impacting a natural phenomenon during a specific time at a particular location (Bansal et al. 2022; Fernandez et al. 2016). Depending on data availability and specific requirements, researchers globally have employed various numerical tools, platforms, and decision criteria to identify, model, and map flood hazard zones in different countries (Mudashiru et al. 2021). In data-scarce areas, multicriteria decision-making (MCDM) methods (Voogd 1982) in conjunction with geographic information systems (GIS) have been widely used due to their capability to handle spatial complexity and uncertainty (de Brito and Evers 2016; Lin et al. 2019; Mudashiru et al. 2021). The MCDM approach also allows participants and stakeholders to integrate their expertise and preferences into decision-making (Pickering and Byrne 2014). Due to such real participatory processes, it has gained widespread acceptance among researchers and decision-makers (de Brito and Evers 2016). Among various MCDM methods, a pairwise comparison approach, particularly Saaty's analytical hierarchy process (AHP) (Saaty 1980), stands out as the most widely adopted method in flood hazard mapping, due to its simplicity and flexibility (Das et al. 2022; Gigović et al. 2017; Hazarika et al. 2018; Lin et al. 2020; Memon et al. 2020; Ramkar and Yadav 2021; Sinha et al. 2008; Stefanidis and Stathis 2013; Vojtek et al. 2021). However, several studies have highlighted the significance of accurately recognizing causative and vulnerable factors for developing reliable flood hazard maps (Jeffers 2013; Kabenge et al. 2017; Mangukiya et al. 2024; Mangukiya and Sharma 2022). Although MCDM methods offer several advantages in this regard by incorporating expert knowledge for selecting causative factors and determining their relative importance, the outcomes may be impacted by the preferences of dominant stakeholders and the uncertainties surrounding their

[1]Research Scholar, Dept. of Hydrology, Indian Institute of Technology Roorkee, Roorkee, Uttarakhand 247667, India. ORCID: https://orcid.org/0000-0001-8132-6958

[2]Research Scientist, Dept. of Geography and Environment, Univ. of Alabama, Tuscaloosa, AL 35487.

[3]Senior Scientist and Head, Water Resources Division, Dept. of Space, North Eastern Space Applications Centre, Umiam, Meghalaya 793122, India.

[4]Scientist and Engineer, National Remote Sensing Center, Indian Space Research Organisation, Hyderabad, Telangana 500037, India.

[5]Assistant Professor, Dept. of Hydrology, Indian Institute of Technology Roorkee, Roorkee, Uttarakhand 247667, India (corresponding author). ORCID: https://orcid.org/0000-0003-2613-2821. Email: ashutosh.sharma@hy.iitr.ac.in

Note. This manuscript was submitted on February 27, 2024; approved on October 14, 2025; published online on December 24, 2025. Discussion period open until May 24, 2026; separate discussions must be submitted for individual papers. This paper is part of the ***Natural Hazards Review***, © ASCE, ISSN 1527-6988.

judgments (Hsu and Sandford 2007), posing a significant open issue.

The recent advancements in machine learning (ML), coupled with geospatial techniques, have ushered in a new paradigm of methods and approaches for flood hazard identification based on the correlation between causative factors and historical flood events (Rehman et al. 2019). Various ML algorithms, such as decision trees, genetic algorithms, gradient boosting algorithms, maximum entropy theories, support vector machines, Naïve Bayes, quick unbiased efficient statistical trees, neural networks, and random forest, have been employed for developing flood hazard maps (Mudashiru et al. 2021). These algorithms are effective because they can accurately capture the complex, nonlinear relationships among flood causative factors and flood hazard categories (Antzoulatos et al. 2022; Darabi et al. 2019; Eini et al. 2020; Mangukiya and Sharma 2022; Norallahi and Seyed Kaboli 2021; Satarzadeh et al. 2022; Singha et al. 2022). While ML algorithms can be more efficient than traditional methods for various tasks, including flood hazard mapping, they also have potential drawbacks such as interpretability, generalization, and explainability (Kumar et al. 2023; Mudashiru et al. 2021; Pradhan et al. 2023). These models are often considered "black boxes" due to a lack of transparency in their decision-making processes (Pradhan et al. 2023). Several studies have utilized explainable and interpretable ML tools to understand how specific outcomes are achieved by the model (Aydin and Iban 2023; Liu et al. 2024; Mangukiya and Sharma 2024; Pradhan et al. 2023; Singha et al. 2024). However, despite the potential benefits, ML techniques can introduce or exacerbate biases in flood hazard management (Antzoulatos et al. 2022). This concern is compounded by the reduced role of human judgment and expertise in the modeling process, where critical decisions are often left to ML algorithms that may lack contextual understanding (Wagenaar et al. 2020). Therefore, it is crucial to implement an integrated approach that combines the strengths of reliable traditional methods with the precision of ML techniques, ensuring that human oversight is integral to the modeling process (Dikshit et al. 2021; Wagenaar et al. 2020).

Among various MCDM methods involving human expertise, AHP (Saaty 1980) has been a widely adopted method for flood hazard and risk mapping (Das et al. 2022; Lin et al. 2020; Memon et al. 2020; Ramkar and Yadav 2021; Vojtek et al. 2021). Among various ML algorithms utilizing machine intelligence, random forest (RF) (Breiman 2001), with its high tolerance for noise and outliers, outperforms the conventional ML algorithms in terms of efficiency, robustness, and computational accuracy (Liu et al. 2021; Mangukiya and Sharma 2022). Several studies have encouraged its use, especially in flood hazard and susceptibility mapping (Antzoulatos et al. 2022; Chen et al. 2020; Deroliya et al. 2022; Feng et al. 2015; Mangukiya and Sharma 2022; Wang et al. 2015). Therefore, this study proposes an integrated MCDM and ML framework using the AHP and random forest algorithm (AHP-RF). We selected the upper Assam region of India for the case study, as this region is highly flood-prone, and flood hazard zonation maps are unavailable to the concerned authorities. Eleven flood causative factors, derived from remote sensing and field-measured data, are used in the analysis to generate flood hazard maps classified into very low to very high categories. The weights of the factors were initially assigned based on Saaty's importance scale (Rahmati et al. 2016), and a piecewise comparison matrix was developed. The normalized initial weights were subsequently adjusted using the feature importance derived from a random forest model to generate the region's high-resolution flood hazard map. For validation, the flood inventory map is used in conjunction with the predicted map from the proposed approach.

Study Area and Datasets

Study Area

Assam state, located in northeastern India, is the second most flood-prone state in the country because of its complex river system and geographical location (Borah et al. 2018; Gupta and Dixit 2022). The flood-prone zones cover approximately 39.58% of the state's total area, including districts such as Dhemaji, Dibrugarh, Lakhimpur, and Tinsukia, which are recurrently afflicted by moderate to high flood situations, affecting a substantial population. In 2022 alone, nearly 34,000 individuals in Dhemaji, Dibrugarh, and Lakhimpur were affected by floods, primarily triggered by incessant rainfall in the proximity to the Brahmaputra River and its major tributaries. The upper Assam districts, namely, Dhemaji and Lakhimpur on the northern bank and Tinsukia and Dibrugarh on the southern bank, are spread over 12,685 km^2 and are home to more than 4.68 million people. These regions suffer from floods and erosion, exacerbated by major tributaries such as Subansiri, Buridihing, GaiNadi, Kumotiya, Jiadhol, Dibang, and Lohit. In light of these challenges, it is essential to develop and adopt efficient strategies for flood mitigation and reducing potential risk in flood-susceptible regions (Demski et al. 2017). Therefore, the upper Assam region, covering four major districts, is selected as a study area and detailed in Fig. 1.

Datasets

The degree of severity, susceptibility, and occurrence of flooding depends on the geographical features, land characteristics, and hydrological and meteorological factors of the region. Therefore, 11 different flood causative factors representing these characteristics were identified and prepared from the satellite imageries and field visits for mapping flood hazards, as briefly outlined in the subsequent sections. These factors were divided into five classes using the Jenks natural breaks classification method (Jenks 1967), as shown in Table S1. Each class of factors was then assigned to a hazard category based on its functional relationship with flooding, logical reasoning, expert judgment, and a literature survey.

Geographical and Topographical Factors

The elevation of a region is a crucial factor in flood causation, with lower elevations posing a higher risk of flooding compared to higher elevations. The digital elevation model (DEM) with a 10-m spatial resolution, representing the absolute height from the CARTOSAT-1 dataset, has been used in this study [Fig. 2(a)]. The CartoDEM was preprocessed to fill sinks and clipped to the study area's boundaries. The slope of a region is another crucial geographical feature influencing the travel time of draining water and the inundation duration of the region. During a flood event, regions with flat slopes are more prone to waterlogging due to the slow speed of water drainage to the outlet, whereas regions with steep slopes are less prone to floods due to quick water drainage. The CartoDEM with a 10-m spatial resolution was utilized to derive the slope of the region using the 'Spatial analyst toolbox' in ArcGIS 10.8 [Fig. 2(b)].

In addition to the river Brahmaputra, its major tributaries also contribute to flooding. Near the confluence points, the probability of flooding is higher; therefore, these locations are considered flood-influencing parameters [Fig. 2(c)]. The drainage network is initially prepared from delineated watersheds, and high-order streams ($\approx$ 3rd order) are identified. The confluence points corresponding to these streams are used in the analysis. Buffer zones are created around the confluence points at distances ranging from 2 to 10 km, and regions near the confluence points are identified as high

Downloaded from ascelibrary.org by UFRPE - UNIVERSIDADE FEDERAL RURAL DE PERNAMBUCO on 09/24/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

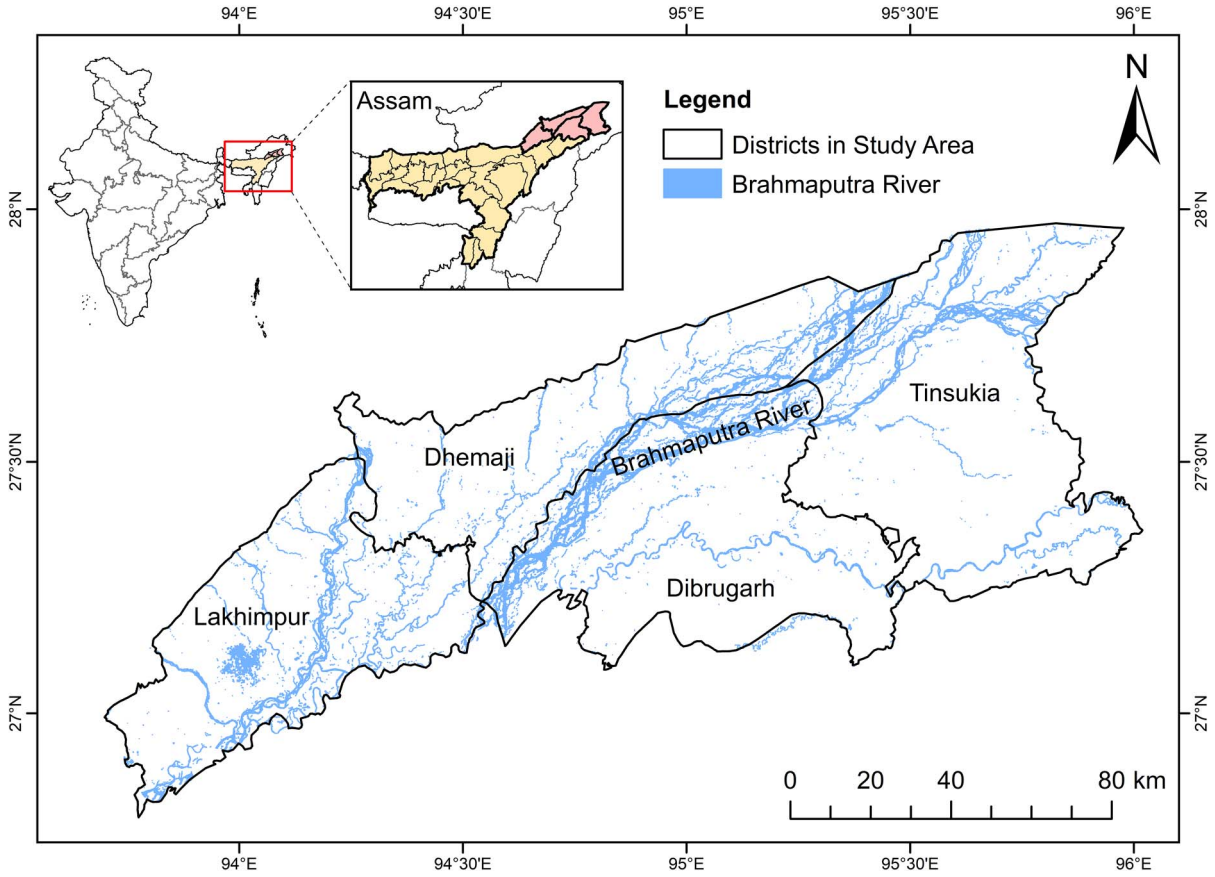


Fig. 1. Location of the upper Assam region illustrating the Brahmaputra River and its tributaries passing through the four major districts.

flood-prone zones. The topographical wetness index (TWI) is commonly used to quantify the generation of runoff and the accumulation of flow at any point within a river basin, accounting for topographical influences. It can be computed based on the contributing drainage area and its slope [Eq. (1)] (Gokceoglu et al. 2005) [Fig. 2(d)]

$$\mathrm{TWI} = \ln\left(\frac{\mathrm{a}}{\tan\theta}\right) \tag{1}$$

where "a" = draining area toward a specific point per unit of contour length; and "θ" = slope in radians.

Land-Use Land Cover Characteristics

The increasing anthropogenic activities modify the land-use land cover (LULC), altering a basin's drainage characteristics and long-term effects on streamflow. A noticeable change has occurred globally in LULC, leading to a progressive rise in the overland flow and finally increasing flood risk (Brody et al. 2014; Sugianto et al. 2022). Here, six land use types are considered, and a hazard category is assigned based on the relative significance of each class [Fig. 2(e)]. For instance, flooding in forest regions does not have much impact on socio-economic or livestock; hence, it is considered a very low category. However, flooding in urban areas causes substantial damage to property and is assigned to a very high hazard category.

Meteorological and Hydrological Factors

Rainfall in alluvial plains is one of the primary reasons for fluvial flooding. The precipitation amount has a direct relation with flooding and is considered a major influencing factor in floods. The rainfall distribution shows a slightly higher trend in the Dhemaji and Lakhimpur districts than in Dibrugarh and Tinsukia. It is evident that regions with higher rainfall are more susceptible to flood hazards than those with lower rainfall. In this study, gridded rainfall data from the India Meteorological Department spanning 30 years (1991–2020) have been used to obtain maximum daily rainfall characteristics of the region [Fig. 2(f)]. Runoff is a crucial hydrological characteristic that indicates the severity of flooding in a particular region. Regions with high runoff are more likely to experience flooding, whereas those with low runoff are less susceptible to flooding. In this study, the SCS curve number (SCS-CN) (Cai and Xu 2022; Katwal et al. 2021) method is used for runoff computations. The runoff response characteristic of a drainage basin is indicated by the curve number (CN). A 10 m resolution LULC map in tandem with the 1:50,000 soil map is used to prepare the curve number for the region. The CN generally varies from 30 to 100, where 30 indicates the lowest potential for runoff and 100 indicates the highest potential for runoff [Fig. 2(g)]. Eqs. (2) and (3) estimate the runoff from the CN and the precipitation data

$$Q = \frac{(P - 0.2S)^2}{P + 0.8S} \tag{2}$$

$$S = 25.4\left(\frac{1000}{\mathrm{CN}} - 10\right) \tag{3}$$

where P = daily precipitation (mm); Q = runoff depth (mm); S = retention parameter; and CN = curve number.

Earthen embankments are highly susceptible to breaches during high-flow events, posing a significant threat to riverine communities.

Downloaded from ascelibrary.org by UFRPE - UNIVERSIDADE FEDERAL RURAL DE PERNAMBUCO on 09/24/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

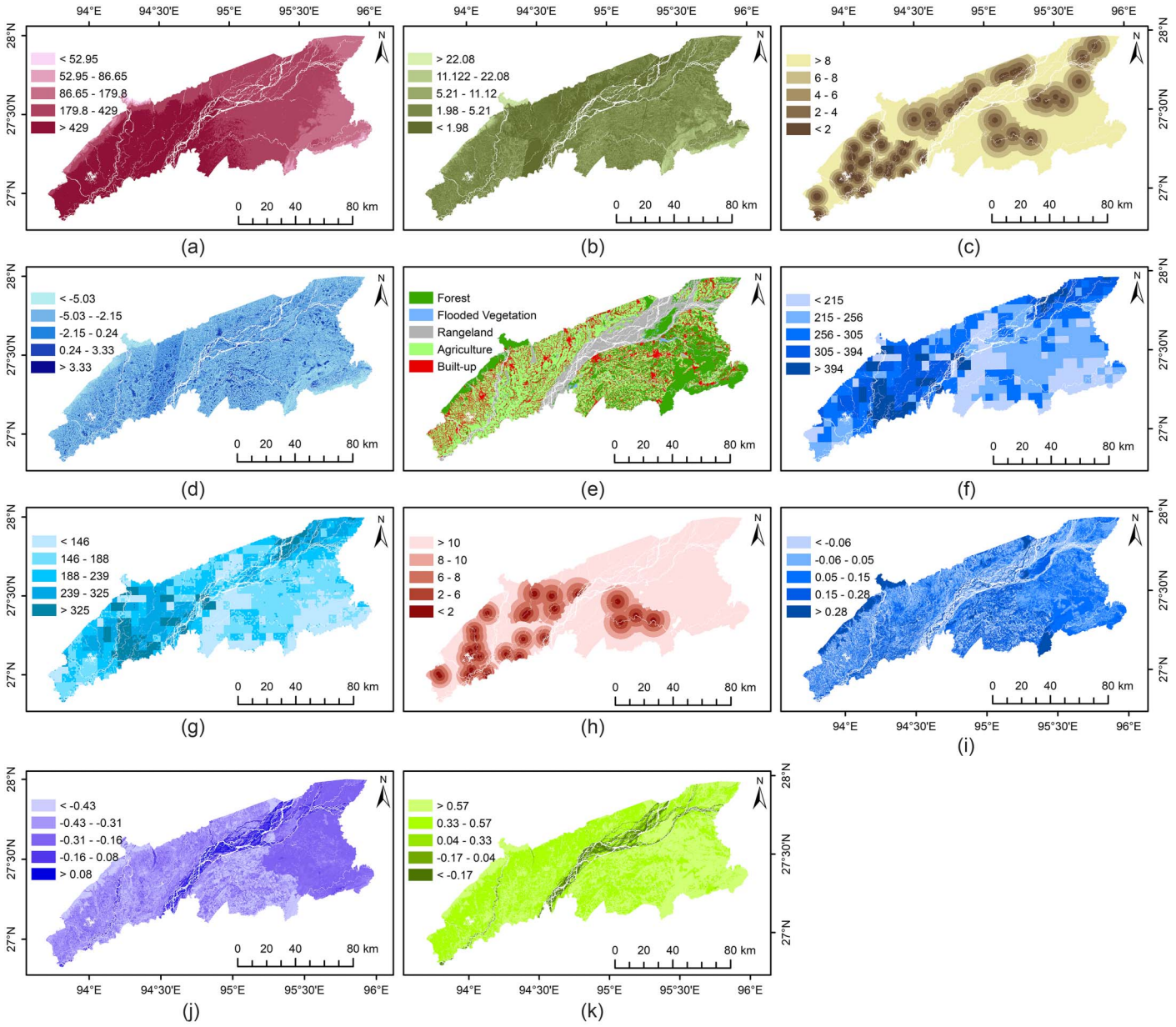


Fig. 2. Flood influencing factors: (a) elevation; (b) slope; (c) distance from the river confluence point; (d) topographic wetness index; (e) land-use land cover; (f) maximum daily rainfall; (g) runoff characteristics; (h) distance from embankment breach; (i) normalized difference moisture index; (j) normalized difference water index; and (k) normalized difference vegetation index.

When embankments are breached, they can lead to widespread inundation of floodplains, making it essential to consider breaches as key flood-influencing factors. In this study, breach points were identified using high-resolution imagery captured by UAVs during field visits, as well as high-resolution satellite imagery obtained after flood events. These points were further validated by the state disaster management authorities. To minimize potential human error in marking the breach locations, buffer zones ranging from 2 to 12 km were created around the breach sites, and flood hazards were systematically assessed based on proximity [Fig. 2(h)].

Remote Sensing Indices

Normalized difference moisture, water, and vegetation index are important flood causative factors. The normalized difference moisture index (NDMI) indicates the water content in the vegetation. Flooding is more common in locations with high NDMI than in regions with lower values. The normalized difference water index (NDWI) is useful in detecting surface water because these surfaces have very low reflectance in the near-infrared (NIR) region and are directly related to floods (Liu et al. 2022). The presence of vegetation is often detected using the normalized difference vegetation index (NDVI), and it tends to have a mitigating effect on floods. This implies that regions with higher NDVI are less prone to flooding, and, conversely, those with lower NDVI values are more susceptible to floods. The study area's NDMI, NDWI, and NDVI have been generated using Sentinel-2 optical imagery of 10 m resolution, as shown in Figs. 2(i–k). The equations used for calculations of NDMI, NDWI, and NDVI are given in Eqs. (4)–(6), respectively

$$\text{NDMI} = \frac{\text{NIR} - \text{SWIR}}{\text{NIR} + \text{SWIR}} \tag{4}$$

$$\text{NDMI} = \frac{\text{Green} - \text{NIR}}{\text{Green} + \text{NIR}} \tag{5}$$

Downloaded from ascelibrary.org by UFRPE - UNIVERSIDADE FEDERAL RURAL DE PERNAMBUCO on 09/24/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

$$\text{NDVI} = \frac{\text{NIR} - \text{RED}}{\text{NIR} + \text{RED}} \quad (6)$$

where NIR = near-infrared band; SWIR = shortwave infrared band; and green and red are the respective color bands.

Methodology

The primary objective of this study is to assess the likelihood and extent of flood hazards in the selected region of upper Assam. This is achieved through the application of the AHP and the proposed integrated AHP-RF approach. Initially, the weights of factors influencing floods were determined based on expert opinions using the AHP method (Baruah et al. 2023; Ramkar and Yadav 2021; Saaty 1980). These AHP-derived weights, combined with flood-influencing factors, were then used to develop a flood hazard map. Subsequently, this map and flood-influencing factors were used as predictors, with the flood inventory map serving as the predictand, for an ensemble ML algorithm, specifically the RF model. The RF model was used to derive the feature importance of flood influencing factors and the AHP flood hazard map. The feature importance of the AHP flood hazard map was then redistributed to the factors based on the weights initially obtained through AHP. Finally, the adjusted weights were used to generate a high-resolution flood hazard map. Fig. 3 provides a schematic depiction of the methodology employed in this study.

Analytical Hierarchy Process for Flood Hazard Mapping

The AHP, developed by Thomas Saaty, is a straightforward yet efficient MCDM model suitable for data-scarce regions (Saaty 1980). In the AHP, the importance intensity for each parameter is assigned based on Saaty's scale of importance to create the piecewise comparison matrix (Melese and Belay 2022; Sinha et al. 2008). The subsequent step involves determining the relative weights of each criterion after completing the weight allocation process. Subsequently, normalized weights (bij) [Eq. (7)], relative weights [Eq. (8)], and the total of normalized weights [Eq. (9)] are simultaneously computed (Baruah et al. 2023). The consistency ratio (CR) is then computed using Eq. (10). It is essential to highlight that the AHP model is considered consistent if the CR is less than or equal to 0.1 (Aksha et al. 2020); otherwise, revisions to the weights are necessary (Chen et al. 2015). More comprehensive information on the AHP can be referred from various literature

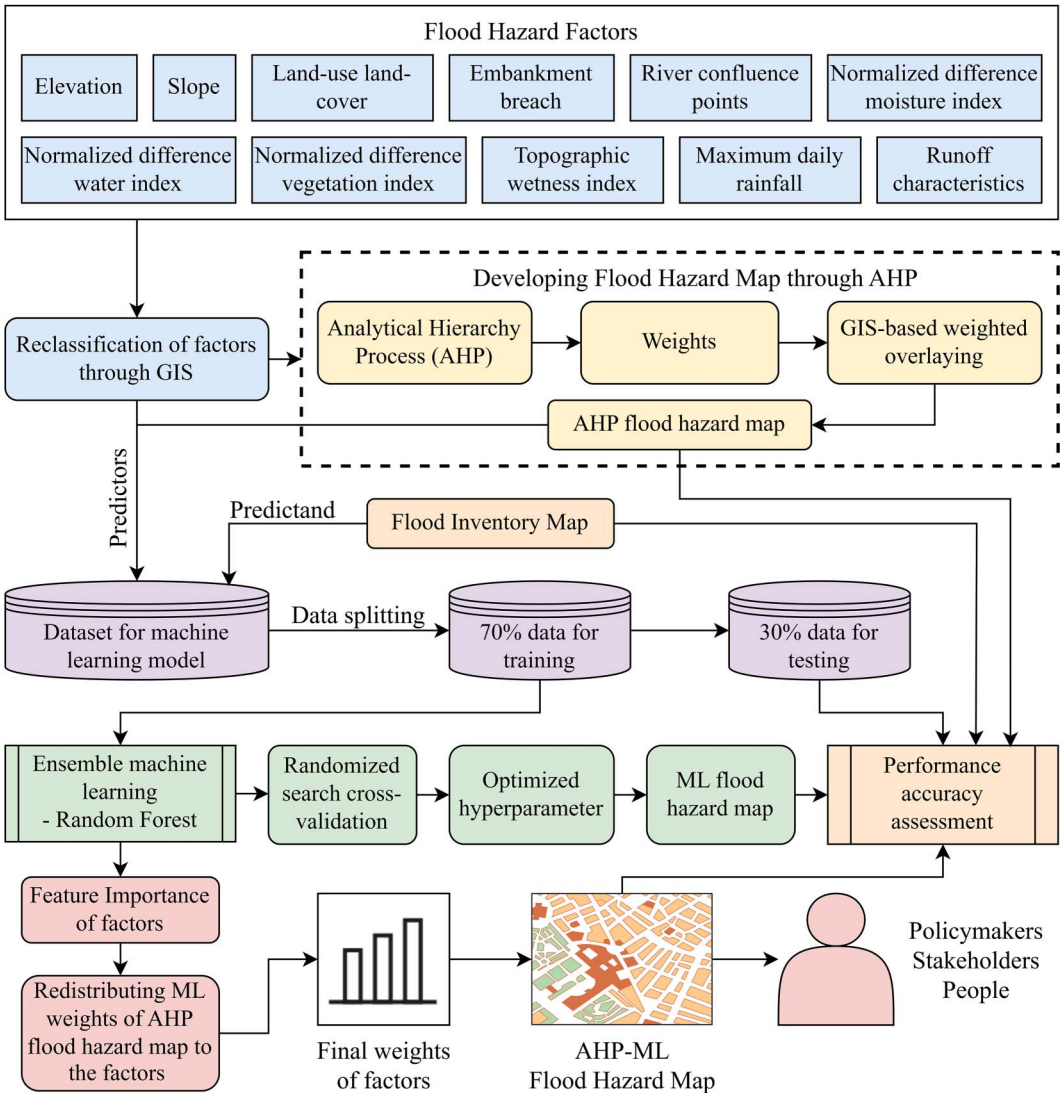


Fig. 3. Schematic depiction of the proposed integrated AHP-machine learning approach for flood hazard mapping.

Downloaded from ascelibrary.org by UFRPE - UNIVERSIDADE FEDERAL RURAL DE PERNAMBUCO on 09/24/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

(de Brito et al. 2018; Cabrera and Lee 2020; Ramkar and Yadav 2021; Wu et al. 2022). In this study, we used the AHP method to derive the initial weights of the factors and prepared a flood hazard map using a GIS-based weighted overlay of the factors

$$b_{ij} = \frac{a_{ij}}{\sum_{i=1}^{n} a_{ij}} \tag{7}$$

$$w_i = \frac{\sum_{j=1}^{n} b_{ij}}{\sum_{i=1}^{n}\sum_{j=1}^{n} b_{ij}}; \quad i = 1, 2, 3 \ldots n \tag{8}$$

$$\sum_{i=1}^{n} w \cdot w_i(x, y) = 1 \tag{9}$$

$$\mathrm{CR} = \frac{\mathrm{CI}}{\mathrm{RI}}; \quad \text{where CI} = \frac{\lambda_{\max} - z}{z - 1} \tag{10}$$

where a_{ij} = individual layers of the piecewise comparison matrix; b_{ij} = normalized weights; w_i = relative weights; RI = random index; CI = consistency index; $\lambda_{\max}$ = maximum eigenvalue of the matrix; and z = number of elements.

Random Forest

The ensemble ML algorithm, especially RF (Breiman 2001), is a widely utilized supervised ML algorithm known for its effectiveness in solving various classification and regression problems. For optimization, the RF model incorporates heuristics and crucial hyperparameters. It serves as a decision tree (DT) application of bagging. RF uses a bootstrap sampling technique to extract random samples of predetermined sizes from the data in order to build a DT model (Mangukiya and Sharma 2022). Following the generation of a DT model from one random sample, the data are reintroduced, ensuring the potential for each random sample's reuse in developing additional decision trees. This approach introduces variability in each DT model within the forest. The outcomes are derived by averaging results from all individual DT models within the forest, effectively reducing variance. Eq. (11) can be used to find the variance of the result if the number of DT models are all identically distributed, have a variance of σ^2, and have a positive pairwise correlation of ρ. (Mangukiya and Sharma 2024). Random selection of $m < p$ input attributes reduces the correlation amongst DT models (ρ) in the RF without considerably increasing variance (σ^2). The RF predictor is then obtained by applying Eq. (12). In this study, the flood hazard map developed through AHP and flood causative factors were used as predictors in the RF model, with the flood inventory map as the predictand. The RandomizedSearchCV method was used to adjust the RF model's hyperparameters (Mangukiya and Sharma 2022). The RF model provided feature importance for all predictors, including the AHP-derived flood hazard map. Subsequently, the feature importance of the AHP-derived flood hazard map was redistributed to the originally used flood-influencing factors in a weighted manner. This redistribution was performed by proportionally allocating the feature importance of the flood hazard map to each factor based on their initial AHP-derived weights. As a result, adjusted weights for each factor were calculated and then used to generate a high-resolution flood hazard map

$$\rho\sigma^2 + \frac{1 - \sigma}{N}\sigma^2 \tag{11}$$

$$\bar{f}_{\mathrm{RF}}^{N}(x) = \frac{1}{N}\sum_{n=1}^{N} T(x; \theta_n) \tag{12}$$

where θ_n = characteristics of the nth DT in terms of the tree depth and minimum samples required in the node and to split the node; and N = total number of DT. More comprehensive information on RF can be referred from various literature (Mangukiya and Sharma 2022, 2024).

Results and Discussion

Flood Hazard Map

The severity of flood hazards depends on flood-influencing factors. The importance of individual flood-influencing layers is always relative to each other, particularly rainfall, elevation, slope, land use, and embankment breaches. In the initial assessment, each flood-influencing factor is weighted using Saaty's scale of importance, and the piecewise comparison matrix is developed as detailed in Table 1. The normalized weights of the flood hazard elements are derived from the comparison matrix. The greatest eigenvalue is then found by multiplying the correlation matrix by the normalized values after the normalized matrix table rows have been averaged (Baruah et al. 2023).

The random index (RI) is a consistency index average whose value is based on the order of the matrix (Baruah et al. 2023). In this study, the RI and CR are determined as 1.51 and 0.077, respectively. To achieve the spatial distribution of flood hazard, each flood-influencing factor was normalized into five categories, as shown in Table S1. Subsequently, the normalized rank for every flood-influencing layer is multiplied by the AHP weights

Table 1. Piecewise comparison matrix for flood hazard assessment of the AHP approach

Flood-influencing factor	DEM	NDMI	Slope	NDWI	Rainfall	Runoff	Embankment Breach	River confluence	NDVI	TWI	Land use
DEM	1	7	3	7	3	5	5	5	7	7	5
NDMI	0.142	1	3	2	0.142	0.142	0.2	0.7	0.33	0.11	0.2
Slope	0.33	5	1	4	3	3	7	7	5	5	5
NDWI	0.042	2	0.2	1	0.142	0.1	0.22	0.33	0.5	0.33	0.2
Rainfall	0.23	7	0.33	5	1	3	5	5	5	7	6
Runoff	0.2	6	0.33	5	0.33	1	7	7	5	5	3
Embankment Breach	0.2	5	0.142	3	0.2	0.142	1	3	3	3	3
River Confluence	0.2	4	0.142	3	0.2	0.142	0.33	1	0.33	3	2
NDVI	0.142	2	0.2	2	0.071	0.2	0.22	2	1	0.22	0.2
TWI	0.142	2	0.1	2	0.071	0.142	0.22	0.33	2	1	3
Land use	0.2	5	0.1	4	0.142	0.1	0.33	0.5	3	0.33	1
Total	2.828	46	8.544	38	8.298	12.968	26.52	31.86	32.16	31.99	28.6

Downloaded from ascelibrary.org by UFRPE - UNIVERSIDADE FEDERAL RURAL DE PERNAMBUCO on 09/24/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

to determine the composite hazard score. The weights used in the flood hazard mapping from the AHP approach are given in Eqs. (13)

$$\begin{aligned}\text{Flood Hazard Zone (FHZ)} &= 0.249 * \text{DEM} + 0.049 * \text{NDMI} + 0.185 * \text{Slope} \\ &+ 0.023 * \text{NDWI} + 0.017 * \text{Rainfall} + 0.128 * \text{Runoff} \\ &+ 0.066 * \text{Embankment breach} + 0.047 * \text{River confluence} \\ &+ 0.034 * \text{NDVI} + 0.045 * \text{TWI} + 0.055 * \text{Landuse} \end{aligned} \quad (13)$$

The flood hazard map is created using the AHP weights for the study area and is shown in Fig. 4(a). The generated flood hazard map from the AHP approach and the flood-influencing factors were utilized as predictors to the RF model, with the flood inventory map by the National Remote Sensing Centre (NRSC) as the predictand. The feature importance from the trained RF model was obtained for all predictors, including the AHP flood hazard map. Subsequently, the feature importance of the AHP flood hazard map was redistributed to factors based on the AHP-derived weights. Finally, the adjusted weights were used to generate a high-resolution flood hazard map, as shown in Fig. 4(b). The results indicate that the highly elevated regions in the Dhemaji, Tinsukia, Dibrugarh, and Lakhimpur show very low hazard categories. Since these high grounds have a steep slope, the possibility of water accumulation is significantly less, thus less risk of flooding. Moreover, most of these regions are forest lands and receive moderate rainfall, reducing the probability of flooding. On the other hand, the valley portions of these regions indicate moderate to high flood hazard zones. Specifically, very high hazard zones are identified near the breach locations of the Dhemaji district. Flood has a negative impact on land-use land cover classes. For instance, during a flood event, the built-up areas and crop fields are more susceptible to damage than forest

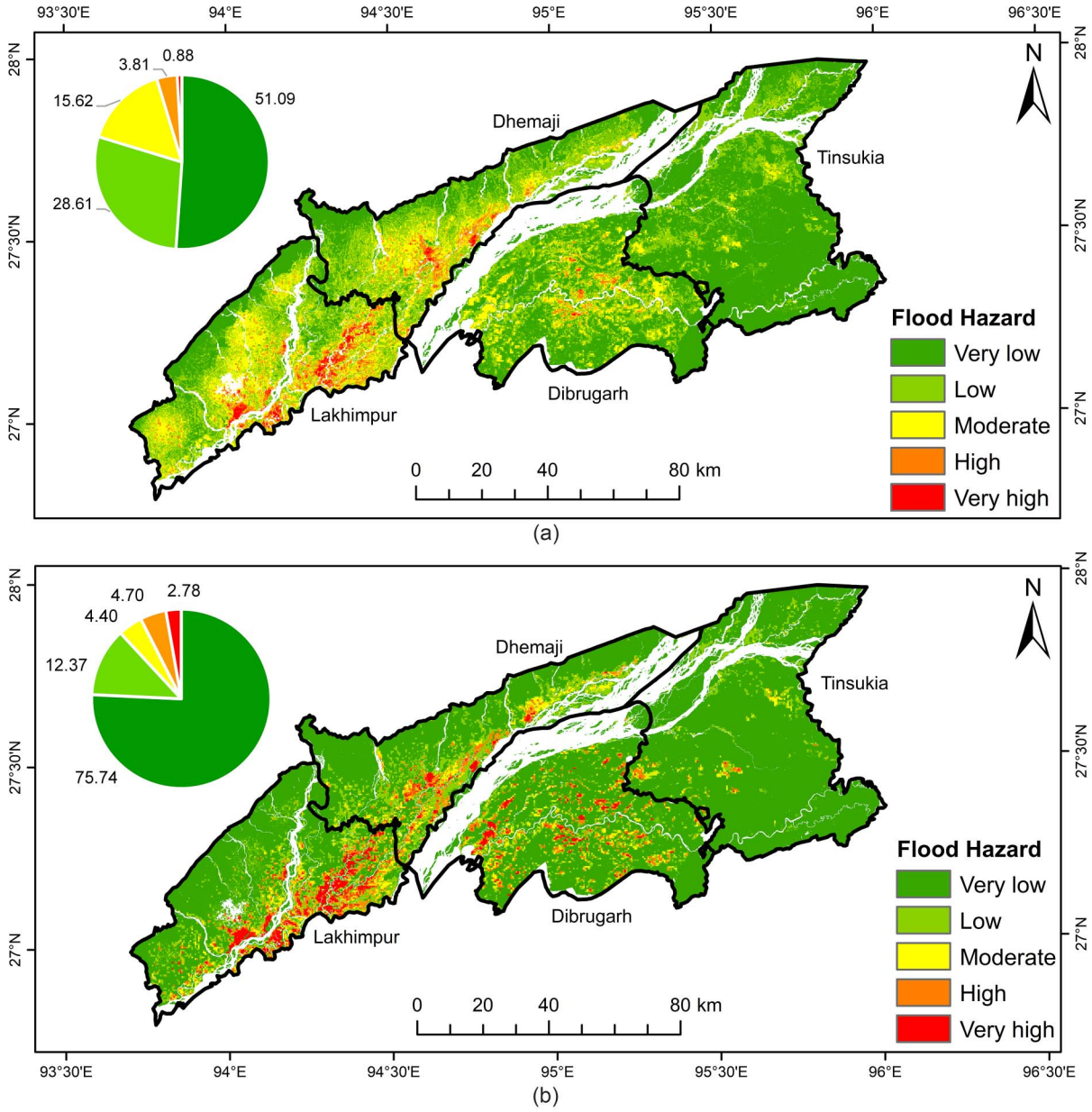


Fig. 4. Flood hazard map was developed using (a) AHP approach; and (b) proposed AHP-RF approach. The pie charts in the figure show the percentage of area under different hazard zones.

Downloaded from ascelibrary.org by UFRPE - UNIVERSIDADE FEDERAL RURAL DE PERNAMBUCO on 09/24/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

Table 2. Areas obtained from the proposed integrated AHP-RF approach under each flood hazard category for the respective district of Assam

District	Flood hazard area (km^2)					Affected LULC class
	Very low	Low	Moderate	High	Very high	
Dhemaji	1,536.14	525.28	219.59	223.10	132.89	Cropland and built-up areas are highly affected
Lakhimpur	1,618.87	492.64	224.37	264.89	255.21	Built-up areas and agricultural land
Dibrugarh	2,066.94	245.35	366.19	374.43	292.11	Built-up areas
Tinsukia	3,270.87	277.94	137.99	117.57	96.62	Agricultural, built up

areas. The affected areas and the land uses are quantified at different hazard zones and presented in Table 2. Out of the four districts, Dibrugarh has more than 292 km^2 area in a very high flood hazard zone, followed by Lakhimpur, Dhemaji, and Tinsukia. Most of the population in these districts are farmers; hence, cropland damage due to flood impacts their socioeconomic condition and livelihood.

Validation and Comparative Analysis of the Generated Flood Hazard Maps

To validate the predicted hazard maps from both the standalone AHP approach and the proposed integrated AHP-RF approach, we utilized the flood inventory map from the NRSC, covering the period from 1998 to 2021 (https://www.nrsc.gov.in/Atlas_FloodHazardZonation). The NRSC flood inventory maps were generated based on the analysis of approximately 215 satellite Microwave SAR datasets, including RISAT-1A, Resourcesat-2, AWiFS, LISS-IV, Sentinel-1A, Radarsat-2, ALOS-2 PALSAR, and THEOS-1. These maps display areas subjected to flooding and the number of occurrences during the analysis period. Additionally, the NRSC flood inventory map was classified into five flood hazard categories based on flooding frequency. *Very high hazard* indicates areas inundated more than 16 times (almost every year), *high hazard* indicates 13 to 15 times, *moderate hazard* indicates nine to 12 times, *low hazard* indicates five to eight times, and *very low hazard* indicates areas inundated one to four times during the analysis period. The NRSC map was selected because it is officially used by the Government of India for identifying flooded regions. For accuracy assessment of the predictions, receiver operating characteristic (ROC) curve analysis is conducted in MATLAB for each hazard category. The ROC curve is a robust metric for evaluating model performance, and the model hypothesis is validated with the area under ROC. Specifically, if the area under the curve (AUC) score is less than or equal to 0.5, the model prediction is deemed unsatisfactory and requires revision. Conversely, the model is considered well-suited for predicting flood patterns if the AUC score is close to 1 (Zou et al. 2007). The findings outlined in Table 3 suggest that the proposed integrated AHP-RF approach outperforms the standalone AHP approach by enhancing the AUC score for each hazard category, particularly for very high hazard zones. It provides reliable results with an overall AUC score of 0.742, compared to 0.654 from the standalone AHP approach. Furthermore, the overall accuracy score of 0.792 from the proposed integrated AHP-RF approach surpasses the accuracy score of 0.595 obtained from the standalone AHP approach.

Table 3. Area under the curve and accuracy score for assessment of flood hazard mapping approaches

AUC score for different flood hazard zones	AHP	AHP-RF
Very low	0.777	0.837
Low	0.645	0.698
Moderate	0.611	0.633
High	0.591	0.672
Very high	0.644	0.871
Overall	0.654	0.742
Accuracy	0.595	0.792

The district-wise comparative analysis of predicted hazard categories from the standalone AHP and the proposed integrated AHP-RF approach reveals an overall trend of overpredicting low and moderate hazard category zones by the standalone AHP approach. In contrast, the proposed integrated AHP-RF approach provides a more accurate representation by adjusting the weights of flood-influencing layers and precisely correcting the hazard category distribution at the district level (Fig. 5). For instance, in the Dhemaji district, the low and moderate hazard category zones were estimated as 950 and 535.2 km^2, respectively, using the AHP approach. In contrast, the proposed integrated AHP-RF approach provided more accurate estimates of 525.3 and 219.6 km^2 for these zones. Similarly, the proposed approach corrected inaccurate predictions from the AHP approach for the Lakhimpur and Dibrugarh districts. For the Tinsukia district, where a majority of the area falls within the very low and low hazard category, both the standalone AHP and proposed integrated AHP-RF approaches performed well. Overall, Table 3 and Fig. 5 suggest that the proposed integrated AHP-RF approach provides more accurate or conservative flood hazard estimations than the standalone AHP approach, demonstrating superior performance in capturing and predicting flood hazard patterns, particularly in very high and high hazard zones. In view of the findings from this study, effective planning and proper mitigation measures can be implemented to benefit the farmers in these regions.

Advantages and Limitations of the Proposed Approach

In this study, runoff is determined using the SCS-CN method based on precipitation and LULC data. While it may seem redundant to include both precipitation and LULC along with runoff, runoff represents a distinct hydrological process that integrates precipitation, LULC, and soil characteristics (Nageswara Rao 2020). This makes it a more accurate representation of surface flow potential, as it reflects the combined effects of soil retention, land use, and rainfall intensity—something that precipitation and LULC alone cannot capture. The integration of the AHP and RF in the proposed AHP-RF approach offers several advantages over the standalone AHP method. Although MCDM methods such as AHP are useful for incorporating expert knowledge and determining the relative importance of causative factors, they are susceptible to the influence of stakeholder preferences and the uncertainties in their judgments, which can introduce bias into the outcomes (Hsu and Sandford 2007). Our results demonstrate that the AHP-RF approach addresses this issue by allowing the data-driven RF model to adjust the factor weights based on their actual contribution to flood hazard prediction, thus reducing the subjectivity inherent in standalone AHP. For instance, the standalone AHP assigns weights

Downloaded from ascelibrary.org by UFRPE - UNIVERSIDADE FEDERAL RURAL DE PERNAMBUCO on 09/24/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

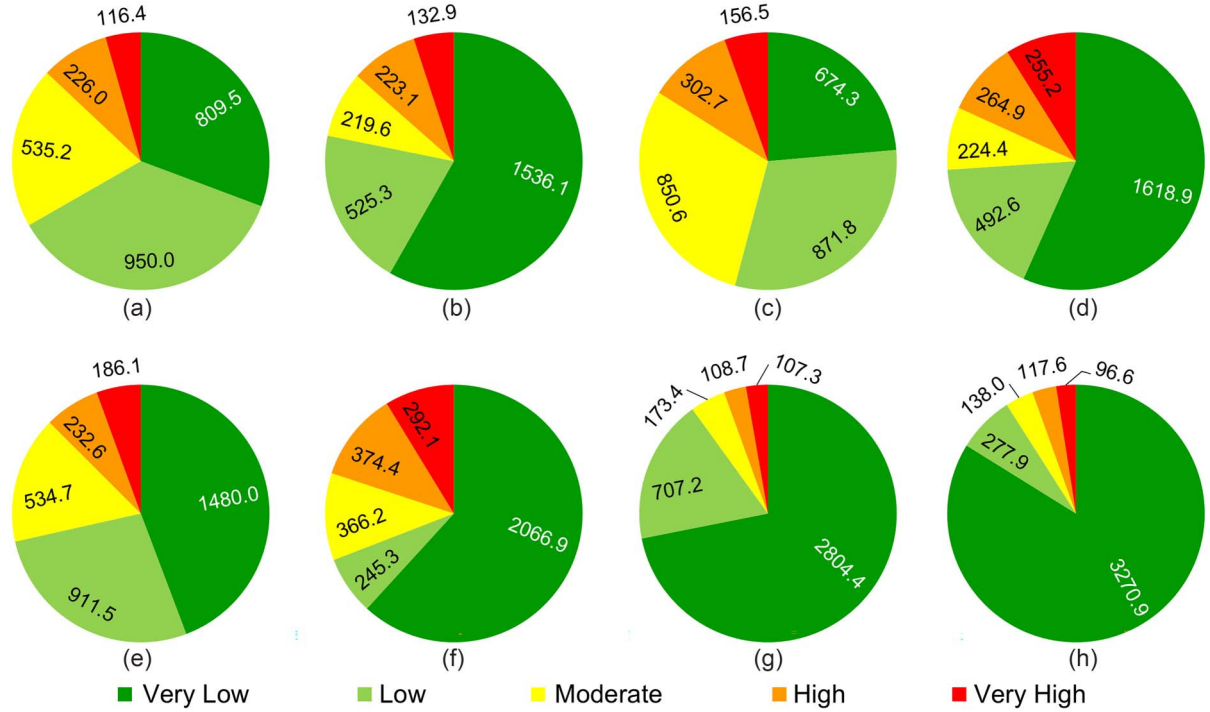


Fig. 5. Flood-affected areas (in km^2) with different hazard zones in each representative district, as obtained from the standalone AHP approach and the proposed integrated AHP-RF approach. (a) Dhemaji – AHP; (b) Dhemaji – AHP-RF; (c) Lakhimpur – AHP; (d) Lakhimpur – AHP-RF; (e) Dibrugarh – AHP; (f) Dibrugarh – AHP-RF; (g) Tinsukia – AHP; and (h) Tinsukia – AHP-RF.

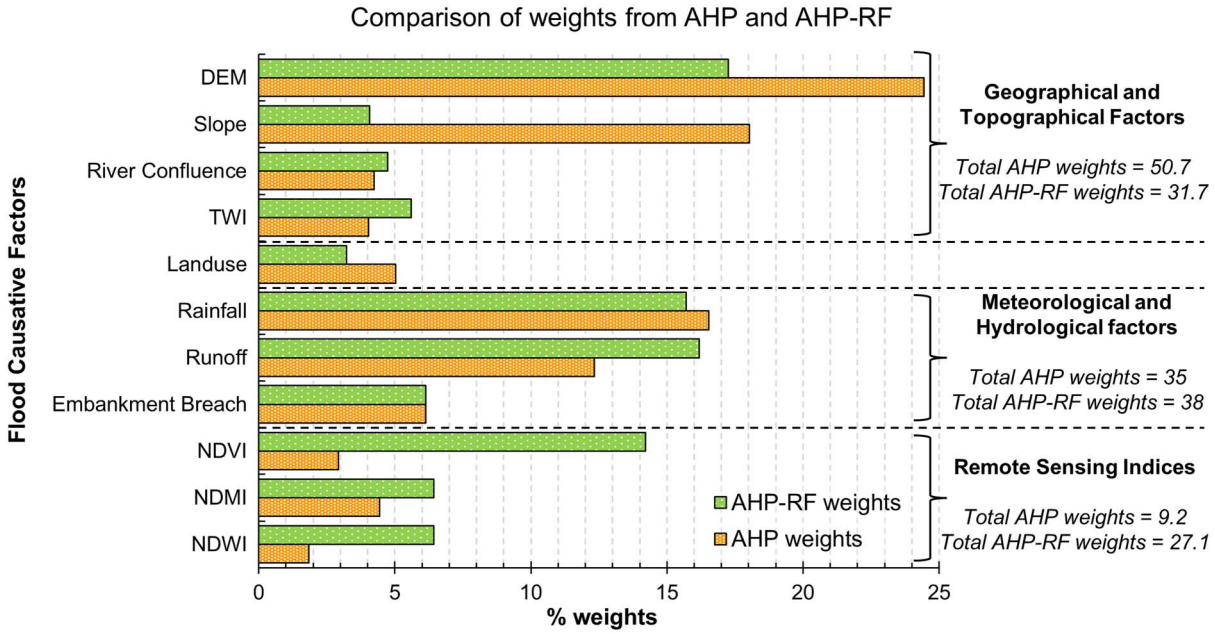


Fig. 6. Comparison of initial weights obtained through AHP and final weights obtained through the proposed AHP-RF approach.

of 16.5, 5, and 12.3 to rainfall, LULC, and runoff, respectively, while the RF model in the AHP-RF approach adjusts these to 15.7, 3.2, and 16.2 (Fig. 6). Furthermore, there is a significant increase in the weight of remote sensing indices from 9.2 in standalone AHP to 27.1 in AHP-RF, highlighting the critical role of satellite-derived data in flood hazard assessment. Conversely, the weights for geographical and topographical factors were reduced from 50.7 in standalone AHP to 31.7 in AHP-RF, reflecting the RF model's ability to balance the importance of various factors more effectively (Fig. 6). By combining the strengths of expert-driven AHP with the precision and data-driven insights of RF, the AHP-RF framework offers a more accurate and balanced flood hazard assessment. This integrated approach not only mitigates the biases introduced by human judgment in AHP but also addresses concerns related to ML algorithms, such as the reduced role of human judgment and expertise in the modeling process. By

Downloaded from ascelibrary.org by UFRPE - UNIVERSIDADE FEDERAL RURAL DE PERNAMBUCO on 09/24/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

leveraging the advantages of both methods, the AHP-RF framework captures intricate patterns and interactions among variables while maintaining human oversight, ensuring a more accurate, transparent, and robust flood hazard assessment.

While the AHP-RF framework offers a balanced integration of expert knowledge and data-driven modeling, it has limitations. One challenge is the reliance on expert judgment in the AHP phase, which can still introduce subjectivity and bias, particularly if stakeholder preferences are not adequately diverse or well-represented (Hsu and Sandford 2007; Munier and Hontoria 2021). Although the proposed approach mitigates some of these biases by adjusting factor weights based on data, the RF model in AHP-RF may be sensitive to the quality and quantity of input data (Picard et al. 2020). In regions where data are sparse, noisy, or lacks spatial resolution, the RF model's performance may degrade. Additionally, the computational complexity of RF can increase with large datasets, potentially making the approach resource-intensive. While the AHP-RF framework improves interpretability compared to standalone ML, the decision-making process in RF still poses some challenges in fully understanding the model's internal workings, especially when communicating results to nontechnical stakeholders.

Conclusions

Flood in alluvial plains is responsible for causing substantial damage to the system and its surroundings. During each monsoon, the flood creates havoc in emerging economies such as India, making it difficult to cope with the resulting losses. With the limited amount of data such as discharge, gauge, and rainfall, it is very difficult to use numerical models and predict the possible hazards. This work proposes an integrated multicriteria decision-making and machine learning framework utilizing the AHP and RF for flood hazard assessment in four vulnerable districts of upper Assam. The state of Assam is highly prone to flood due to its geographical location, presence of a vast river network, and highly varying climatology and topography; hence, based on these characteristics, 11 different layers are considered flood-influencing layers. The initial weights were assigned using AHP depending on the relative importance of each layer, and the flood hazard map was prepared. This flood hazard map and flood-influencing factors were given as input to the RF model to adjust the weights based on feature importance and develop a high-resolution flood hazard map of the region. For validation, the flood occurrence map is used in the study. The comparative analysis of the standalone AHP and the proposed integrated AHP-RF approach suggests that the proposed integrated AHP-RF approach provides more accurate or conservative flood hazard estimations than the standalone AHP approach, demonstrating superior performance in capturing and predicting flood hazard patterns, particularly in very high and high hazard zones. The results indicate that built-up and agricultural lands are highly affected by floods among different land use classes. The district-wise analysis shows that more than 250 km^2 of the area falls under the very high hazard category in Dibrugarh and Lakhimpur, followed by 100 km^2 in Dhemaji and Tinsukia. To our knowledge, the flood hazard map developed in this study, using several hydrometeorological and geospatial inputs in the GIS platform for the upper Assam region, is the first-ever high-resolution flood hazard map available for the region. The generated map will be valuable for disaster management agencies to prepare action plans as well as mitigation measures for the highly affected areas. The insights from this research can also be applied to other basin areas with different climate conditions and topographical features to assess flood hazards.

Data Availability Statement

The data used to support the findings of this study are available from the corresponding author upon request.

Author Contributions

Nikunj K. Mangukiya: Conceptualization; Data curation; Formal analysis; Methodology; Software; Visualization; Writing – original draft. Anupal Baruah: Conceptualization; Data curation; Methodology; Supervision; Writing – review and editing. Diganta Barman: Conceptualization; Investigation; Supervision; Writing – review and editing. B. M. Arjun: Conceptualization; Investigation; Writing – review and editing. Ashutosh Sharma: Conceptualization; Funding acquisition; Supervision; Writing – review and editing.

Supplemental Materials

Table S1 is available online in the ASCE Library (www.ascelibrary.org).

References

Aksha, S. K., L. M. Resler, L. Juran, and L. W. Carstensen. 2020. "A geospatial analysis of multi-hazard risk in Dharan, Nepal." *Geomatics Nat. Hazards Risk* 11 (1): 88–111. https://doi.org/10.1080/19475705.2019.1710580.

Ali, H., P. Modi, and V. Mishra. 2019. "Increased flood risk in Indian sub-continent under the warming climate." *Weather Clim. Extremes* 25 (Sep): 100212. https://doi.org/10.1016/j.wace.2019.100212.

Antzoulatos, G., et al. 2022. "Flood hazard and risk mapping by applying an explainable machine learning framework using satellite imagery and GIS data." *Sustainability* 14 (6): 3251. https://doi.org/10.3390/su14063251.

Aydin, H. E., and M. C. Iban. 2023. "Predicting and analyzing flood susceptibility using boosting-based ensemble machine learning algorithms with SHapley additive exPlanations." *Nat. Hazards* 116 (3): 2957–2991. https://doi.org/10.1007/s11069-022-05793-y.

Bansal, N., M. Mukherjee, and A. Gairola. 2022. "Evaluating urban flood hazard index (UFHI) of Dehradun city using GIS and multi-criteria decision analysis." *Model. Earth Syst. Environ.* 8 (3): 4051–4064. https://doi.org/10.1007/s40808-021-01348-5.

Baruah, A., D. Barman, B. M. Arjun, B. L. Chyne, and S. P. Aggarwal. 2023. "Holistic framework for flood hazard assessment in a transboundary basin." *Acta Geophys.* 72 (2): 1017–1032. https://doi.org/10.1007/s11600-023-01063-y.

Beshir, A. A., and J. Song. 2021. "Urbanization and its impact on flood hazard: The case of Addis Ababa, Ethiopia." *Nat. Hazards* 109 (1): 1167–1190. https://doi.org/10.1007/s11069-021-04873-9.

Borah, S. B., T. Sivasankar, M. N. S. Ramya, and P. L. N. Raju. 2018. "Flood inundation mapping and monitoring in Kaziranga National Park, Assam using Sentinel-1 SAR data." *Environ. Monit. Assess.* 190 (9): 520. https://doi.org/10.1007/s10661-018-6893-y.

Breiman, L. 2001. "Random forests." *Mach. Learn.* 45 (1): 5–32. https://doi.org/10.1023/a:1010933404324.

Brody, S., R. Blessing, A. Sebastian, and P. Bedient. 2014. "Examining the impact of land use/land cover characteristics on flood losses." *J. Environ. Plann. Manage.* 57 (8): 1252–1265. https://doi.org/10.1080/09640568.2013.802228.

Cabrera, J. S., and H. S. Lee. 2020. "Flood risk assessment for Davao Oriental in the Philippines using geographic information system-based multi-criteria analysis and the maximum entropy model." *J. Flood Risk Manage.* 13 (2): e12607. https://doi.org/10.1111/jfr3.12607.

Cai, X., and D. Xu. 2022. "Simulation and optimization strategy of storm flood safety pattern based on SCS-CN model." *Int. J. Environ. Res. Public Health* 19 (2): 698. https://doi.org/10.3390/ijerph19020698.

Downloaded from ascelibrary.org by UFRPE - UNIVERSIDADE FEDERAL RURAL DE PERNAMBUCO on 09/24/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

Downloaded from ascelibrary.org by UFRPE - UNIVERSIDADE FEDERAL RURAL DE PERNAMBUCO on 09/24/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

Chen, W., et al. 2020. "Modeling flood susceptibility using data-driven approaches of naïve Bayes tree, alternating decision tree, and random forest methods." *Sci. Total Environ.* 701 (Jan): 134979. https://doi.org/10.1016/j.scitotenv.2019.134979.

Chen, Y., R. Liu, D. Barrett, L. Gao, M. Zhou, L. Renzullo, and I. Emelyanova. 2015. "A spatial assessment framework for evaluating flood risk under extreme climates." *Sci. Total Environ.* 538 (Dec): 512–523. https://doi.org/10.1016/j.scitotenv.2015.08.094.

Darabi, H., B. Choubin, O. Rahmati, A. Torabi Haghighi, B. Pradhan, and B. Kløve. 2019. "Urban flood risk mapping using the GARP and QUEST models: A comparative study of machine learning techniques." *J. Hydrol.* 569 (Feb): 142–154. https://doi.org/10.1016/j.jhydrol.2018.12.002.

Das, S., S. Sarkar, and D. P. Kanungo. 2022. "GIS-based landslide susceptibility zonation mapping using the analytic hierarchy process (AHP) method in parts of Kalimpong Region of Darjeeling Himalaya." *Environ. Monit. Assess.* 194 (4): 234. https://doi.org/10.1007/s10661-022-09851-7.

de Brito, M. M., and M. Evers. 2016. "Multi-criteria decision-making for flood risk management: A survey of the current state of the art." *Nat. Hazards Earth Syst. Sci.* 16 (4): 1019–1033. https://doi.org/10.5194/nhess-16-1019-2016.

de Brito, M. M., M. Evers, and A. D. S. Almoradie. 2018. "Participatory flood vulnerability assessment: A multi-criteria approach." *Hydrol. Earth Syst. Sci.* 22 (1): 373–390. https://doi.org/10.5194/hess-22-373-2018.

Demski, C., S. Capstick, N. Pidgeon, R. G. Sposato, and A. Spence. 2017. "Experience of extreme weather affects climate change mitigation and adaptation responses." *Clim. Change* 140 (2): 149–164. https://doi.org/10.1007/s10584-016-1837-4.

Deroliya, P., M. Ghosh, M. P. Mohanty, S. Ghosh, K. H. V. D. Rao, and S. Karmakar. 2022. "A novel flood risk mapping approach with machine learning considering geomorphic and socio-economic vulnerability dimensions." *Sci. Total Environ.* 851 (Dec): 158002. https://doi.org/10.1016/j.scitotenv.2022.158002.

Dikshit, A., B. Pradhan, and A. M. Alamri. 2021. "Pathways and challenges of the application of artificial intelligence to geohazards modelling." *Gondwana Res.* 100 (Dec): 290–301. https://doi.org/10.1016/j.gr.2020.08.007.

Eini, M., H. S. Kaboli, M. Rashidian, and H. Hedayat. 2020. "Hazard and vulnerability in urban flood risk mapping: Machine learning techniques and considering the role of urban districts." *Int. J. Disaster Risk Reduct.* 50 (Nov): 101687. https://doi.org/10.1016/j.ijdrr.2020.101687.

Feng, Q., J. Liu, and J. Gong. 2015. "Urban flood mapping based on unmanned aerial vehicle remote sensing and random forest classifier—A case of Yuyao, China." *Water* 7 (4): 1437–1455. https://doi.org/10.3390/w7041437.

Fernandez, P., S. Mourato, and M. Moreira. 2016. "Social vulnerability assessment of flood risk using GIS-based multicriteria decision analysis. A case study of Vila Nova de Gaia (Portugal)." *Geomatics Nat. Hazards Risk* 7 (4): 1367–1389. https://doi.org/10.1080/19475705.2015.1052021.

Gigović, L., D. Pamučar, Z. Bajić, and S. Drobnjak. 2017. "Application of GIS-interval rough AHP methodology for flood hazard mapping in urban areas." *Water* 9 (6): 360. https://doi.org/10.3390/w9060360.

Gokceoglu, C., H. Sonmez, H. A. Nefeslioglu, T. Y. Duman, and T. Can. 2005. "The 17 March 2005 Kuzulu landslide (Sivas, Turkey) and landslide-susceptibility map of its near vicinity." *Eng. Geol.* 81 (1): 65–83. https://doi.org/10.1016/j.enggeo.2005.07.011.

Gupta, L., and J. Dixit. 2022. "A GIS-based flood risk mapping of Assam, India, using the MCDA-AHP approach at the regional and administrative level." *Geocarto Int.* 37 (26): 11867–11899. https://doi.org/10.1080/10106049.2022.2060329.

Hazarika, N., D. Barman, A. K. Das, A. K. Sarma, and S. B. Borah. 2018. "Assessing and mapping flood hazard, vulnerability and risk in the Upper Brahmaputra River valley using stakeholders' knowledge and multicriteria evaluation (MCE)." *J. Flood Risk Manage.* 11 (Feb): S700–S716. https://doi.org/10.1111/jfr3.12237.

Hsu, C. C., and B. A. Sandford. 2007. "The Delphi technique: Making sense of consensus." *Pract. Assess. Res. Eval.* 12 (1): 10. https://doi.org/10.7275/pdz9-th90.

Jeffers, J. M. 2013. "Integrating vulnerability analysis and risk assessment in flood loss mitigation: An evaluation of barriers and challenges based on evidence from Ireland." *Appl. Geogr.* 37 (1): 44–51. https://doi.org/10.1016/j.apgeog.2012.10.011.

Jenks, G. F. 1967. "The data model concept in statistical mapping." In Vol. 7 of *International yearbook of cartography*. London: George Philip & Son.

Kabenge, M., J. Elaru, H. Wang, and F. Li. 2017. "Characterizing flood hazard risk in data-scarce areas, using a remote sensing and GIS-based flood hazard index." *Nat. Hazards* 89 (3): 1369–1387. https://doi.org/10.1007/s11069-017-3024-y.

Katwal, R., J. Li, T. Zhang, C. Hu, M. A. Rafique, and Y. Zheng. 2021. "Event-based and continous flood modeling in Zijinguan watershed, Northern China." *Nat. Hazards* 108 (1): 733–753. https://doi.org/10.1007/s11069-021-04703-y.

Kumar, V., H. M. Azamathulla, K. V. Sharma, D. J. Mehta, and K. T. Maharaj. 2023. "The state of the art in deep learning applications, challenges, and future prospects: A comprehensive review of flood forecasting and management." *Sustainability* 15 (13): 10543. https://doi.org/10.3390/su151310543.

Lin, K., H. Chen, C.-Y. Xu, P. Yan, T. Lan, Z. Liu, and C. Dong. 2020. "Assessment of flash flood risk based on improved analytic hierarchy process method and integrated maximum likelihood clustering algorithm." *J. Hydrol.* 584 (May): 124696. https://doi.org/10.1016/j.jhydrol.2020.124696.

Lin, L., Z. Wu, and Q. Liang. 2019. "Urban flood susceptibility analysis using a GIS-based multi-criteria analysis framework." *Nat. Hazards* 97 (2): 455–475. https://doi.org/10.1007/s11069-019-03615-2.

Liu, Y., X. Lu, Y. Yao, N. Wang, Y. Guo, C. Ji, and J. Xu. 2021. "Mapping the risk zoning of storm flood disaster based on heterogeneous data and a machine learning algorithm in Xinjiang, China." *J. Flood Risk Manage.* 14 (1): e12671. https://doi.org/10.1111/jfr3.12671.

Liu, Y., D. Xiao, and W. Yang. 2022. "An algorithm for early rice area mapping from satellite remote sensing data in southwestern Guangdong in China based on feature optimization and random Forest." *Ecol. Inf.* 72 (Dec): 101853. https://doi.org/10.1016/j.ecoinf.2022.101853.

Liu, Z., T. Felton, and A. Mostafavi. 2024. "Interpretable machine learning for predicting urban flash flood hotspots using intertwined land and built-environment features." *Comput. Environ. Urban Syst.* 110 (Jun): 102096. https://doi.org/10.1016/j.compenvurbsys.2024.102096.

Mangukiya, N. K., S. Kushwaha, and A. Sharma. 2024. "A novel multi-model ensemble framework for fluvial flood inundation mapping." *Environ. Modell. Software* 180 (Sep): 106163. https://doi.org/10.1016/j.envsoft.2024.106163.

Mangukiya, N. K., and A. Sharma. 2022. "Flood risk mapping for the lower Narmada basin in India: A machine learning and IoT-based framework." *Nat. Hazards* 113 (2): 1285–1304. https://doi.org/10.1007/s11069-022-05347-2.

Mangukiya, N. K., and A. Sharma. 2024. "Alternate pathway for regional flood frequency analysis in data-sparse region." *J. Hydrol.* 629 (Feb): 130635. https://doi.org/10.1016/j.jhydrol.2024.130635.

Mangukiya, N. K., and S. M. Yadav. 2022. "Integrating 1D and 2D hydrodynamic models for semi-arid river basin flood simulation." *Int. J. Hydrol. Sci. Technol.* 14 (2): 206–228. https://doi.org/10.1504/IJHST.2022.124549.

Melese, T., and T. Belay. 2022. "Groundwater potential zone mapping using analytical hierarchy process and GIS in Muga watershed, Abay Basin, Ethiopia." *Global Challenges* 6 (1): 2100068. https://doi.org/10.1002/gch2.202100068.

Memon, N., D. P. Patel, N. Bhatt, and S. B. Patel. 2020. "Integrated framework for flood relief package (FRP) allocation in semiarid region: A case of Rel River flood, Gujarat, India." *Nat. Hazards* 100 (1): 279–311. https://doi.org/10.1007/s11069-019-03812-z.

Mohanty, M. P., S. Mudgil, and S. Karmakar. 2020. "Flood management in India: A focussed review on the current status and future challenges." *Int. J. Disaster Risk Reduct.* 49 (Oct): 101660. https://doi.org/10.1016/j.ijdrr.2020.101660.

Mohapatra, P. K., and R. D. Singh. 2003. "Flood management in India." *Nat. Hazards* 28 (1): 131–143. https://doi.org/10.1023/A:1021178000374.
Mudashiru, R. B., N. Sabtu, I. Abustan, and W. Balogun. 2021. "Flood hazard mapping methods: A review." *J. Hydrol.* 603 (Dec): 126846. https://doi.org/10.1016/j.jhydrol.2021.126846.
Munier, N., and E. Hontoria. 2021. "Uses and limitations of the AHP method." In *Management for professionals*. Cham, Switzerland: Springer.
Nageswara Rao, K. 2020. "Analysis of surface runoff potential in ungauged basin using basin parameters and SCS-CN method." *Appl. Water Sci.* 10 (1): 47. https://doi.org/10.1007/s13201-019-1129-z.
Norallahi, M., and H. Seyed Kaboli. 2021. "Urban flood hazard mapping using machine learning models: GARP, RF, MaxEnt and NB." *Nat. Hazards* 106 (1): 119–137. https://doi.org/10.1007/s11069-020-04453-3.
Picard, S., C. Chapdelaine, C. Cappi, L. Gardes, E. Jenn, B. Lefevre, and T. Soumarmon. 2020. "Ensuring dataset quality for machine learning certification." In *Proc., 2020 IEEE Int. Symp. on Software Reliability Engineering Workshops (ISSREW)*, 275–282. New York: IEEE.
Pickering, C., and J. Byrne. 2014. "The benefits of publishing systematic quantitative literature reviews for PhD candidates and other early-career researchers." *Higher Educ. Res. Dev.* 33 (3): 534–548. https://doi.org/10.1080/07294360.2013.841651.
Pradhan, B., S. Lee, A. Dikshit, and H. Kim. 2023. "Spatial flood susceptibility mapping using an explainable artificial intelligence (XAI) model." *Geosci. Front.* 14 (6): 101625. https://doi.org/10.1016/j.gsf.2023.101625.
Rahmati, O., A. Haghizadeh, and S. Stefanidis. 2016. "Assessing the accuracy of GIS-based analytical hierarchy process for watershed prioritization; Gorganrood River Basin, Iran." *Water Resour. Manage.* 30 (3): 1131–1150. https://doi.org/10.1007/s11269-015-1215-4.
Ramkar, P., and S. M. Yadav. 2021. "Flood risk index in data-scarce river basins using the AHP and GIS approach." *Nat. Hazards* 109 (1): 1119–1140. https://doi.org/10.1007/s11069-021-04871-x.
Rehman, S., M. Sahana, H. Hong, H. Sajjad, and B. Bin Ahmed. 2019. "A systematic review on approaches and methods used for flood vulnerability assessment: Framework for future research." *Nat. Hazards* 96 (2): 975–998. https://doi.org/10.1007/s11069-018-03567-z.
Saaty, T. L. 1980. *The analytic hierarchy process: Planning, priority setting, resource allocation*. New York: McGraw-Hill.
Satarzadeh, E., A. Sarraf, H. Hajikandi, and M. S. Sadeghian. 2022. "Flood hazard mapping in western Iran: Assessment of deep learning vis-à-vis machine learning models." *Nat. Hazards* 111 (2): 1355–1373. https://doi.org/10.1007/s11069-021-05098-6.
Singh, O., and M. Kumar. 2017. "Flood occurrences, damages, and management challenges in India: A geographical perspective." *Arabian J. Geosci.* 10 (5): 102. https://doi.org/10.1007/s12517-017-2895-2.
Singha, C., V. K. Rana, Q. B. Pham, D. C. Nguyen, and E. Łupikasza. 2024. "Integrating machine learning and geospatial data analysis for comprehensive flood hazard assessment." *Environ. Sci. Pollut. Res.* 31 (35): 48497–48522. https://doi.org/10.1007/s11356-024-34286-7.
Singha, C., K. C. Swain, M. Meliho, H. G. Abdo, H. Almohamad, and M. Al-Mutiry. 2022. "Spatial analysis of flood hazard zoning map using novel hybrid machine learning technique in Assam, India." *Remote Sens.* 14 (24): 6229. https://doi.org/10.3390/rs14246229.
Sinha, R., G. V. Bapalu, L. K. Singh, and B. Rath. 2008. "Flood risk analysis in the Kosi river basin, north Bihar using multi-parametric approach of analytical hierarchy process (AHP)." *J. Indian Soc. Remote Sens.* 36 (4): 335–349. https://doi.org/10.1007/s12524-008-0034-y.
Stefanidis, S., and D. Stathis. 2013. "Assessment of flood hazard based on natural and anthropogenic factors using analytic hierarchy process (AHP)." *Nat. Hazards* 68 (2): 569–585. https://doi.org/10.1007/s11069-013-0639-5.
Sugianto, S., A. Deli, E. Miswar, M. Rusdi, and M. Irham. 2022. "The effect of land use and land cover changes on flood occurrence in Teunom watershed, Aceh Jaya." *Land* 11 (8): 1271. https://doi.org/10.3390/land11081271.
Thakur, D. A., and M. P. Mohanty. 2023. "A synergistic approach towards understanding flood risks over coastal multi-hazard environments: Appraisal of bivariate flood risk mapping through flood hazard, and socio-economic-cum-physical vulnerability dimensions." *Sci. Total Environ.* 901 (Nov): 166423. https://doi.org/10.1016/j.scitotenv.2023.166423.
Vojtek, M., J. Vojteková, R. Costache, Q. B. Pham, S. Lee, A. Arshad, S. Sahoo, N. T. T. Linh, and D. T. Anh. 2021. "Comparison of multi-criteria-analytical hierarchy process and machine learning-boosted tree models for regional flood susceptibility mapping: A case study from Slovakia." *Geomatics Nat. Hazards Risk* 12 (1): 1153–1180. https://doi.org/10.1080/19475705.2021.1912835.
Voogd, H. 1982. "Multicriteria evaluation with mixed qualitative and quantitative data." *Environ. Plann. B: Plann. Des.* 9 (2): 221–236. https://doi.org/10.1068/b090221.
Wagenaar, D., A. Curran, M. Balbi, A. Bhardwaj, R. Soden, E. Hartato, G. Mestav Sarica, L. Ruangpan, G. Molinario, and D. Lallemant. 2020. "Invited perspectives: How machine learning will change flood risk and impact assessment." *Nat. Hazards Earth Syst. Sci.* 20 (4): 1149–1161. https://doi.org/10.5194/nhess-20-1149-2020.
Wang, Z., C. Lai, X. Chen, B. Yang, S. Zhao, and X. Bai. 2015. "Flood hazard risk assessment model based on random forest." *J. Hydrol.* 527 (Aug): 1130–1141. https://doi.org/10.1016/j.jhydrol.2015.06.008.
Wu, J., X. Chen, and J. Lu. 2022. "Assessment of long and short-term flood risk using the multi-criteria analysis model with the AHP-Entropy method in Poyang Lake basin." *Int. J. Disaster Risk Reduct.* 75 (Jun): 102968. https://doi.org/10.1016/j.ijdrr.2022.102968.
Yadav, S. M., and N. K. Mangukiya. 2021. "Semi-arid river basin flood: Causes, damages, and measures." In *Proc., Fifth Int. Conf. in Ocean Engineering (ICOE 2019)*, 201–212. Singapore: Springer.
Zou, K. H., A. J. O'Malley, and L. Mauri. 2007. "Receiver-operating characteristic analysis for evaluating diagnostic tests and predictive models." *Circulation* 115 (5): 654–657. https://doi.org/10.1161/CIRCULATIONAHA.105.594929.

Downloaded from ascelibrary.org by UFRPE - UNIVERSIDADE FEDERAL RURAL DE PERNAMBUCO on 09/24/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.